



# Prediction of three-phase product distribution and bio-oil heating value of biomass fast pyrolysis based on machine learning

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## ABSTRACT

In this work, by mining the experimental data of fast pyrolysis of lignocellulosic biomass in bubbling fluidized bed in previous literature, regression prediction models were established for three-phase product distribution and bio-oil heating value (HHV) based on gradient boosting, random forest, support vector machine, and multilayer perceptron algorithms. Comprehensive feedstock characteristics and pyrolysis conditions were considered and compared as input features. Among the several algorithms, random forest is most suitable for the prediction of three-phase product yields and bio-oil HHV with the benefits of high accuracy and good generalization ability. Visual analysis of the model shows that pyrolysis temperature is the most critical factor affecting three-phase product distribution, while bio-oil HHV is more affected by the feedstock characteristics such as the contents of C and H. The highest yield and HHV of bio-oil is obtained at about 480 °C, suggesting 480 °C as the optimum pyrolysis temperature of fast pyrolysis of biomass in a bubbling fluidized bed. As for the feedstock characteristics, high contents of C and H and low content of O are favorable to the enhancement of bio-oil HHV, indicating the crucial importance of feedstock pretreatment such as torrefaction to the quality improvement of bio-oil.

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## 1. Introduction

The development and utilization of fossil fuels have promoted the fast growth of human science and technology society. However, it has also brought concerns about the energy supply and environmental pollution. Biomass is the only carbon-based new energy with abundant and renewable reserves, which has been attracting significant attention. Lignocellulosic biomass, as the most abundant biomass resource, including agricultural and forestry wastes, mainly consists of cellulose, hemicellulose, and lignin. It was estimated that the world's annual production of lignocellulosic biomass is about 100 billion tons containing a huge energy content about five times the world's annual energy consumption [1]. Biochemical conversion and thermochemical conversion are

currently the primary utilization methods of biomass energy. Among various thermochemical conversion technologies, a pyrolysis is a promising form and the initial stage of different thermochemical conversion technologies, which can convert solid biomass into biochar, bio-oil, and gas in an oxygen-free or oxygen-deficient environment [2,3].

According to different heating rates and pyrolysis temperatures, pyrolysis can be divided into torrefaction, slow pyrolysis, fast pyrolysis, flash pyrolysis, and high-pressure liquefaction etc., corresponding to different target products [4]. The fast pyrolysis of biomass refers to the pyrolysis conditions of medium pyrolysis temperature (400–500 °C) and short volatiles residence time (0.5–2 s) to obtain bio-oil, which contains low sulfur and benzene and produces 50% less harmful substances when used as fuels compared to fossil fuels [5,6]. Therefore, compared with fossil fuels, bio-oil bears the advantage of being clean and environmentally friendly. However, its low stability, low heating value, high acidity, and high viscosity restrict its utilization, and further costly

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upgrading is often needed. Higher heating value (HHV) is one of the important bio-oil characteristics, which determines the quality of bio-oil as a liquid fuel.

The main factors influencing the product distribution and the HHV of bio-oil from fast pyrolysis of biomass can be classified as feedstock characteristics and pyrolysis conditions. Feedstock characteristics can be represented from different aspects such as the compositions by proximate analysis, the elementary compositions by ultimate analysis, the contents of the three components (cellulose, hemicellulose, and lignin), the particle size, etc., which are not completely independent from each other. Efforts have been made to reveal the correlations between them and the product distribution as well as the bio-oil HHV. It was proposed that the high volatile content of the biomass feedstock makes the pyrolysis reaction easier and promotes the formation of bio-oil products [7]. Instead, pyrolysis of feedstock with high ash content results in low bio-oil yields and high biochar and gas yields [8]. The elementary composition of the feedstock can also affect the pyrolysis product. Chiodo et al. [9] found that pyrolysis feedstock's hydrogen element can increase the HHV of bio-oil. Through pyrolysis of cellulose, lignin, and corn stalks, Gani et al. [10] found that cellulose and hemicellulose promote bio-oil production. In contrast, lignin plays a positive role in the formation of biochar. Besides, the fast pyrolysis of tea waste with a particle size of 0.3 and 1.5 mm produced biochar yields of 15 wt% and 20 wt%, respectively, indicating the promotion effect of larger particle size on biochar [11]. As for pyrolysis conditions, the pyrolysis temperature is one of the most influential factors. For example, when the pyrolysis temperature rises from 300 to 500 °C, the bio-oil yield increases from 26.5 to 36.6 wt%, HHV increased from 10.53 to 13.41 MJ/kg, gas yield increased from 23.2 to 37.4 wt% [12].

Besides the aforementioned work, extensive experimental and simulation work has also been done to explore the effects of two kinds of factors on pyrolysis and propose a widely applicable prediction model for reactor design and optimization [4,13]. However, the detailed correlations are still obscure, and it remains challenging to establish a detailed reaction-transportation model of biomass fast pyrolysis, which is also too time-consuming for reactor-scale simulation. The numerical modeling simulation time for rapid pyrolysis of biomass is generally ranged from 50 to 200 s [14–17]. The simplified explicit models, such as global or semi-global kinetic models, detailed lumped models, network models, etc., are often fitted with limited data and are not suitable for variable feedstock characteristics and pyrolysis conditions. Furthermore, in the available models, biomass is always regarded as a whole or a simple sum of the three components. Some other influential factors, such as the possible interactions between the three components, inorganic species, and extractives are still always ignored [13], despite some recent papers trying to address the issues [18–20]. In addition, the contents of the three components are often used as the input characteristics of feedstock in the modeling of biomass pyrolysis. Instead, info of biomass proximate and ultimate analysis is always discarded or just used to estimate the contents of the three components [18], considering the high correlations between different aspects of feedstock characteristics in virgin biomass. Together with the discard or indirect use of proximate and ultimate analysis, which contain additional information about inorganic species, extractives, etc., the inaccuracy also be introduced to the prediction for biomass pyrolysis. It is known that inorganic substances in biomass not only affect the yield of the product but also affect the composition of the product [13,21]. Furthermore, a consideration of extractives in kinetic modeling also proved beneficial to prediction.

In this context, without the need for a prior explicit model, machine learning provides options for coping with the complexity.

It can build predictive models with limited cognition and mine data's inherent related information. Cao et al. [22] constructed a prediction model of biochar yield for cow manure-based pyrolysis based on artificial neural networks (ANN) and support vector machines (SVM). Also, using SVM and ANN algorithms, Chen et al. [23] constructed a model for the prediction of three-phase product distribution and bio-oil HHV of biomass pyrolysis. The results show that SVM performed higher accuracy than ANN. Zhu et al. [24] used random forest (RF) to build a model to predict the production of slow pyrolysis biochar and biochar's carbon content. It was found that the accuracy of the model is improved by the addition of the elementary compositions to the collection of the pyrolysis conditions and the three components compositions in biomass as the input features. Pathy et al. [25] constructed an extreme gradient boosting regression (XGB) model using feedstock characteristics and pyrolysis conditions as inputs and successfully predicted the yield of biochar obtained by algal biomass pyrolysis. The research results show that the model inputting both ultimate composition and proximate analysis composition is more accurate than the model constructed by only inputting ultimate composition. Tang et al. [26] constructed a pyrolysis bio-oil yield and bio-oil H content prediction model based on multiple linear regression (MLR) and RF. The results show that the prediction of the RF model effect is more evident than that of the MLR model.

Despite the utilization of machine learning for biomass pyrolysis mentioned above, different feedstock characteristics and pyrolysis conditions have not been comprehensively considered. Furthermore, there is a lack of discussion on models constructed by different machine learning algorithms and different input feature combinations. A proper combination of different aspects of feedstock characteristics is expected to benefit the accuracy of prediction together with an appropriate machine learning algorithm. In this work, proximate analysis, ultimate analysis, the contents of the three components, and the particle size were all considered in the prediction of the yields of biochar, bio-oil, and synthesis gases and the HHV of bio-oil for biomass fast pyrolysis in bubbling fluidized bed. Considering the multivariate nonlinear process of biomass pyrolysis, several learning algorithms, i.e., RF, SVM, XGB, and multilayer perceptron (MLP), were adopted and compared. The models were optimized from two aspects: the selection of input features and the adjustment of hyperparameters. The relative importance of biomass characteristics and pyrolysis conditions in the pyrolysis process was studied. Exploring the internal information of the model helps to deepen the understanding of the fast pyrolysis mechanism, which provides engineering guidance for regulating the yield of pyrolysis products and the improvement of bio-oil performance.

## 2. Methods

### 2.1. Sample data collection and processing

Sample data were excavated from the fast pyrolysis experiment in bubbling fluidized beds reported in the literature. The data is obtained directly from the table or extracted from the picture using g3data software. Considering the wide range of feedstock sources, the difference, and the reproducibility of experimental conditions, some data were screened to construct and test the model. The feedstock characteristics and pyrolysis conditions in each set of data after screening are different, and each collection of data contains the practical values of all variables. 122 sets of data were used for biochar, bio-oil, and gas prediction [27–41]. 92 sets of data are used for the HHV of bio-oil prediction [39–54].

All biomass used in the pyrolysis experiments is lignocellulosic biomass, including wheat straw, pine wood, corn cob, eucalyptus,

etc. The feedstock samples were dried, ground into smaller biomass particles, and subjected to proximate analysis, ultimate analysis, and structural analysis to obtain the feedstock characteristics. The fast pyrolysis experiments were carried out under inert conditions with nitrogen as the carrier gas and preheating to the preset pyrolysis temperature. Different pyrolysis conditions were provided by changing the feed rate, pyrolysis temperature, and gas flow rate. After the pyrolysis, the biochar, bio-oil, and gases were obtained through condensation, separator, and other devices. All the values from proximate analysis, ultimate analysis, structural analysis, product mass fraction, and HHV of bio-oil are presented on a dry basis.

The characteristics of feedstock and pyrolysis conditions and the type of reactor will affect the pyrolysis of biomass. The key factors affecting the distribution of biomass pyrolysis products can be divided into five categories. The components of biomass by proximate analysis are ash (Ash), volatile (V), and fixed carbon (FC) content. The variables Ash and V-FC are recorded as input I and II, respectively. The three components of biomass are cellulose (Cel), hemicellulose (Hem), and lignin (Lig). Without considering the interaction of the three components, biomass pyrolysis can be regarded as the superposition of the three components pyrolysis, so the content of the three components determines the product yield. Their variable contents are recorded as input III (Cel-Hem-Lig). Element compositions are the element contents of carbon (C), hydrogen (H), oxygen (O), and nitrogen (N), and the input variables C-H-O-N are recorded as input IV. The content of each element in the feedstock affects the content of each element in the pyrolysis product, and the element content of the product has an important impact on the characteristics of the product. Biomass particle size (PS,  $\mu\text{m}$ ) is registered as input V, which can affect the yield of pyrolysis products by affecting the heat and mass transfer in the pyrolysis process. Pyrolysis conditions include pyrolysis temperature (HT,  $^{\circ}\text{C}$ ), GFR (GFR,  $\text{m}^3/\text{h}$ ), and FR (FR,  $\text{kg}/\text{h}$ ). The pyrolysis temperature is one of the most critical parameters in the pyrolysis process, and it has an important influence on the product yield and product characteristics. GFR is the volume of gas entering the pyrolysis reactor per unit time, and FR is the mass of biomass fed into the reactor per unit time. GFR and FR mainly affect the distribution and properties of pyrolysis products by affecting the heat and mass transfer in the pyrolysis process. The input variables HT-GFR-FR are recorded as input VI. The inputs I, II, III, IV, V, and VI contain fourteen input variables. The four output variables are the yields of biochar (char), bio-oil (oil), and gas and the HHV of bio-oil (HHV,  $\text{MJ}/\text{kg}$ ).

All data used to build the model was standardized using Z-Score before training and testing. The standardized formula is formula (1):

$$x_i^z = \frac{(x_i - \mu)}{\delta} \quad (1)$$

Where  $x_i^z$  is the normalized value of each input feature,  $x_i$  is the original value of the input feature,  $\mu$  and  $\delta$  are the averages and standard deviation of each input feature, respectively.

## 2.2. Input feature selection

To prevent the model from overfitting, reduce the model's complexity and find the optimal prediction model, different combinations of feedstock input characteristics are selected for model training. Ash (I) is present in all the combinations representing the inorganic part of the biomass. As for the organic part, considering the overlap of V-FC (II), Cel-Hem-Lig (III), and C-H-O-N (IV), seven combinations (C1 3 + C2 3 + C3 3) were determined together with

PS (V) and HT-GFR-FR (VI), as shown in Table 1.

## 2.3. Training models

### 2.3.1. XGB model

XGB is an integrated algorithm to minimize residual error. It has the advantages of fast iteration speed, sound effects, and the ability to handle large-scale data. The basis of XGB is the gradient boosting algorithm, which is an integrated algorithm for building predictive models [55]. The integrated algorithm makes multiple weak evaluators on the data and aggregates all weak evaluators' modeling results to obtain better regression or classification performance than a single model. The rough modeling process is shown in Fig. 1 [56]. Firstly, a gradient boosting tree is established as a weak estimator, followed by a gradual iteration. The iteration reduces deviation and generalization error as the criterion, and one tree is added during each iteration, gradually forming a substantial estimator integrated with many tree models.

Each regression tree's basic unit is the leaf node, which has a prediction score, also known as the leaf weight representing all the samples on this leaf node. The tree's regression value is indicated by  $f_k(x_i)$ , where  $k$  represents the  $k$ -th decision tree and  $x_i$  represents the feature vector corresponding to sample  $i$ . The regression result of the ensemble model is the sum of the prediction scores of all trees. Assuming that there are  $K$  decision trees in the ensemble model, the prediction result given by the model on sample  $i$  is:

$$\hat{y}_i^{(k)} = \sum_k^K f_k(x_i) \quad (2)$$

The establishment of all models is done through the Python machine learning library (Scikit-learn). The filtered data set is divided into a training set and a test set with a ratio of 4:1. The training set is used to build the model, and the test set is used to test the accuracy of the model. The essential parameters of the XGB model are the number of weak evaluators in the `n_estimators` integration and the sample ratio drawn when a subsample is randomly sampled. The learning rate in the `learning_rate` integration, which is used to control the iteration rate, prevent the model from overfitting, and control the gamma model's complexity [57]. The parameters of the model were adjusted by cross-validation (4-fold) and grid search, and the adjusted scoring standard is the regression coefficient ( $R^2$ ).

### 2.3.2. RF model

RF is a data analysis method based on decision trees, which is used to process the nonlinear relationship between variables applied to classification or regression problems. Compared to other machine learning algorithms, it has a very high accuracy rate, can process input samples with higher dimensional features, and has lower data requirements [58]. The implementation of RF is divided into the following three sub-steps. Firstly, the Bootstrap method is used to randomly select  $m$  samples from the original set with

**Table 1**  
Model variable input information.

| No. | Input combination     | Input variable                                         |
|-----|-----------------------|--------------------------------------------------------|
| 1   | I, II, III, IV, V, VI | Ash, V, FC, Cel, Hem, Lig, C, H, O, N, PS, HT, GFR, FR |
| 2   | I, III, IV, V, VI     | Ash, Cel, Hem, Lig, C, H, O, N, PS, HT, GFR, FR        |
| 3   | I, II, IV, V, VI      | Ash, V, FC, C, H, O, N, PS, HT, GFR, FR                |
| 4   | I, II, III, V, VI     | Ash, V, FC, Cel, Hem, Lig, PS, HT, GFR, FR             |
| 5   | I, II, V, VI          | Ash, V, FC, PS, HT, GFR, FR                            |
| 6   | I, III, V, VI         | Ash, Cel, Hem, Lig, PS, HT, GFR, FR                    |
| 7   | I, IV, V, VI          | Ash, C, H, O, N, PS, HT, GFR, FR                       |

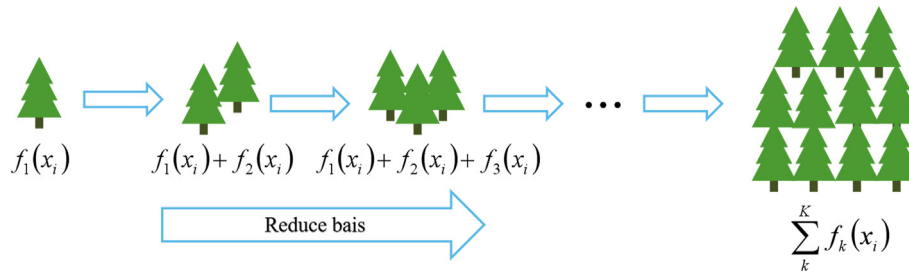


Fig. 1. Schematic diagram of the gradient lifting method modeling process.

replacement sampling and perform a total of  $n$  samplings to generate  $n$  training sets. Secondly, the developed  $n$  training is used to training  $n$  decision tree models, respectively. Thirdly, the generated multiple decision trees form a random forest, and the average of the predicted values of numerous trees determine the final prediction result. The key hyperparameters of the RF model are the number of decision trees ( $n\_estimators$ ), the number of features considered when looking for the best split point ( $max\_features$ ), the maximum depth of the tree ( $max\_depth$ ), the minimum number of samples required to split internal nodes ( $min\_samples\_split$ ) and the minimum number of samples needed on leaf nodes ( $min\_samples\_leaf$ ) [59]. The best hyperparameters were obtained through grid search and cross-validation (4-fold), and the adjusted scoring standard is the  $R^2$ .

### 2.3.3. SVM model

SVM is based on statistical theory and the principle of structural risk minimization. It has obvious advantages in dealing with nonlinear and high-dimensional problems [60]. Its basic model is defined as the linear classifier with the most considerable feature space interval. The learning strategy is to maximize the break, which can finally be transformed into the solution of a convex quadratic programming problem. For a given training sample  $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_i, y_i)\}$ ,  $y_i \in R$ , the SVM algorithm hopes to train a regression model such that  $f(x)$  and  $y$  as close as possible. To solve the nonlinear regression problem, the data needs to be projected from the original space  $x$  to the new space  $\Phi(x)$  [61].  $\Phi$  is a mapping function representing an inevitable non-linear transformation;  $w$  and  $b$  are the model parameters to be determined  $f(x)$ . The expression is shown in equation (3):

$$f(x) = w^T \Phi(x) + b \quad (3)$$

The parameters  $w$  and  $b$  are the weight matrix and the bias vector, respectively. Values of  $w$  and  $b$  are determined by the regularized minimum loss function expression (4).

$$\min_{w,b,\epsilon} \frac{\|w\|^2}{2} + C \sum_{i=1}^n \epsilon_i \quad (4)$$

The constraints of (4) are:

$$y_i[w\Phi(x_i) + b] \geq 1 - \epsilon_i, i = 1, 2, \dots, n \quad (5)$$

Here,  $\epsilon (\epsilon_i \geq 0)$  is the relaxation variable introduced by relaxing the original discriminant function's unequal conditions, and  $C$  is the regularization constant. To find a suitable mapping function  $\Phi$ , it is necessary to introduce the kernel function (5), which can use the vector calculation in the original space of the data to represent the dot product's result in the space after the dimension is increased.

$$K(x_i, x_j) = \Phi(x_i)^T \cdot \Phi(x_j) \quad (6)$$

The three-state product of fast pyrolysis of biomass is predicted to be a nonlinear problem, so the nonlinear radial basis function (RBF) is selected as the kernel function (6), and  $\delta$  is the bandwidth of the kernel function:

$$K(x_i, x_j) = \exp\left(\frac{-\|x_i - x_j\|^2}{2\delta^2}\right) \quad (7)$$

Under this premise of the kernel function, the model parameters, i.e. model complexity control gamma and regularization constant  $C$ , were adjusted through the learning curve with  $R^2$  as the scoring standard.

### 2.3.4. MLP model

MLP is a kind of artificial neural network. It imitates a biological neural network's structure and function and changes the internal structure based on external information [62]. As shown in Fig. 2, MLP comprises an input layer, hidden layer, and output layer, which have the neuron as the unit. The input layer is composed of input features (i.e., neuron)  $\{x_1, x_2, \dots, x_m\}$ , which set inputs for the hidden layer. Each neuron in the first hidden layer first performs a linear transformation of the value from the input layer:  $w_1x_1 + w_2x_2 + \dots + w_mx_m$ , and then acts on the nonlinear activation function  $f(x)$  to obtain the value of the first hidden layer. The neurons in the other hidden layers linearly transform the previously hidden layer's value and then act on the activation function to obtain the value of the hidden layer. The output layer receives the last hidden layer's values and converts them into output values [63]. The activation function and weight optimization were selected as the rectified linear unit function and adaptive moment estimation (optimizer based on stochastic gradient). The number of hidden layers was selected for two layers. Other parameters include

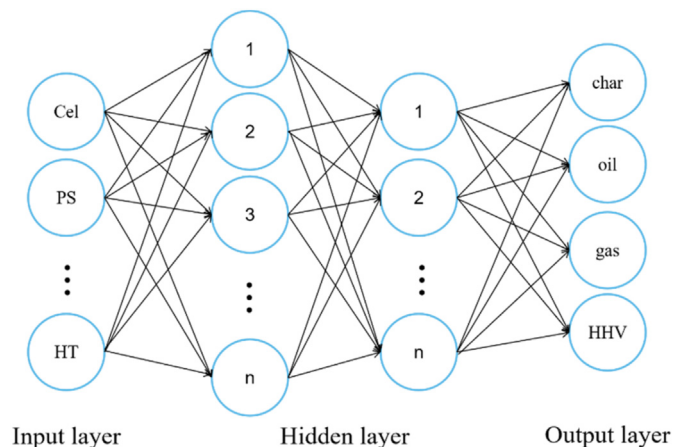


Fig. 2. Neural network structure diagram.



the vector of the number of neurons in each hidden layer `hidden_layer_sizes`, the regularization parameter `alpha`, the initial learning rate `learning_rate_init`, etc., which were obtained through the learning curve, and the scoring standard is  $R^2$ .

## 2.4. Performance evaluation

According to the regression coefficient ( $R^2$ ) and root mean square error (RMSE) of the test data, the model's accuracy was evaluated. The calculation formulas of  $R^2$  and RMSE are shown in the following equations (8) and (9):

$$R^2 = 1 - \frac{\sum_{i=1}^N (Y_i - Y_i^{pred})^2}{\sum_{i=1}^N (Y_i - Y_{ave}^{exp})^2} \quad (8)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - Y_i^{pred})^2} \quad (9)$$

Where  $Y_{exp\ i}$ ,  $Y_{pred\ i}$  and  $Y_{exp\ ave}$  are the experimental value, the model prediction, and the average of experimental values, respectively.

## 3. Results and discussions

### 3.1. Comparison of XGB, RF, SVM, and MLP model

$R^2$  and RMSE constructed by different input feature selections and algorithms are shown in Fig. 3 and Fig. 4, respectively. The RF model and XGB model performed well in predicting biochar yield,

with  $R^2$  greater than 0.80 and RMSE less than 4. The largest  $R^2$  and the smallest RMSE were obtained for the RF model with the No. 6 of input feature selections. As for bio-oil yield, the RF, XGB, and MLP models have high accuracy. The largest  $R^2$  and the smallest RMSE were obtained by No.6 for all models, following the order of RF > MLP > XGB > SVM and SVM > XGB > MLP > RF, respectively. In addition, the RF, MLP, and XGB model performed well in predicting gas yield, and  $R^2$  are all greater than 0.80, and RMSEs are all lower than 4. In the prediction of bio-oil HHV, the RF model has obvious advantages with higher  $R^2$  and low RMSE over other models. Although the RF model does not perform best in all the input feature selections, for example, the XGB model performs best in the prediction of biochar by the No. 7 of input feature selections, but the prediction levels of the two models are very close. In addition, when the input feature selection is No. 6, the random forest model always performs best. In general, the RF algorithm's model has the highest prediction accuracy for the three-phase product yields and the bio-oil HHV. The RF algorithm introduces sample randomness and feature randomness and is an integration of multiple decision trees, which makes the RF less prone to overfitting compared to other algorithms, and perform well on the test set [58].

### 3.2. Comparison of RF models with different input feature selections

The accuracy results of the model built with additional information on the test set are shown in Table 2. In general, the accuracy of the models built by seven different input feature selections is relatively high, indicating their ability to predict the yields of pyrolysis three-phase products and the bio-oil HHV. In predicting the yields of biochar, bio-oil, and gas, No.6 performs best with the highest  $R^2$  and the lowest RMSE, which indicates that the three-component classification of biomass has a stronger relationship

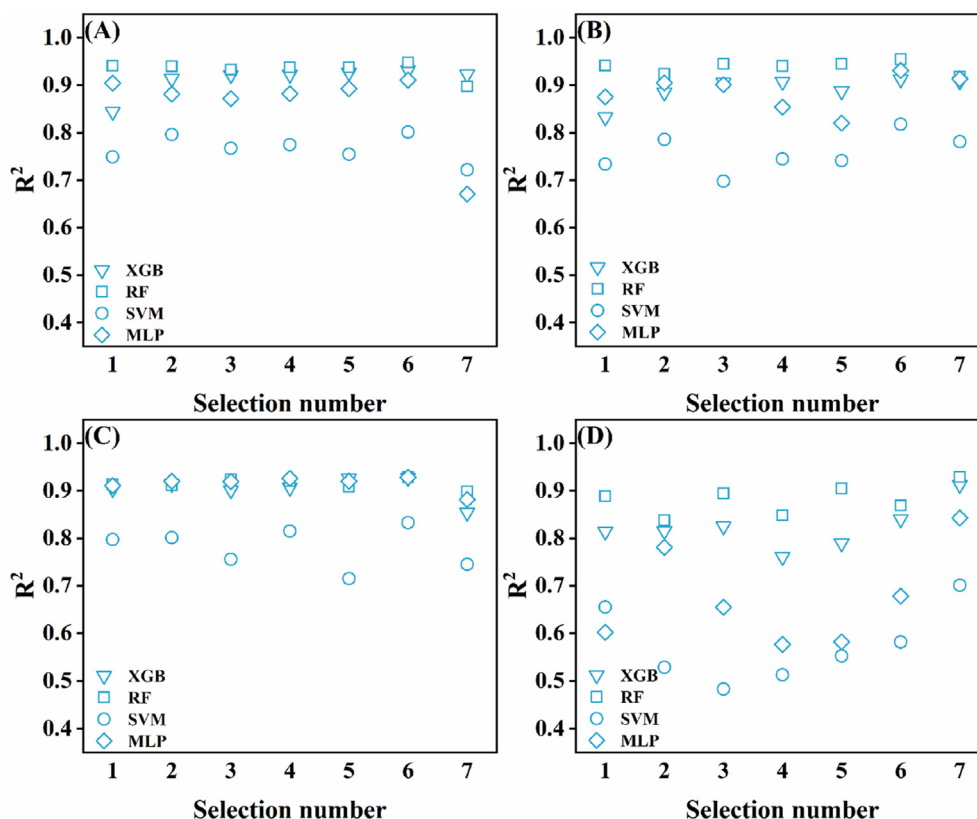


Fig. 3.  $R^2$  of XGB, RF, SVM, and MLP with different input feature selections for the prediction of (A) biochar product, (B) bio-oil product, (C) gas product, and (D) HHV.

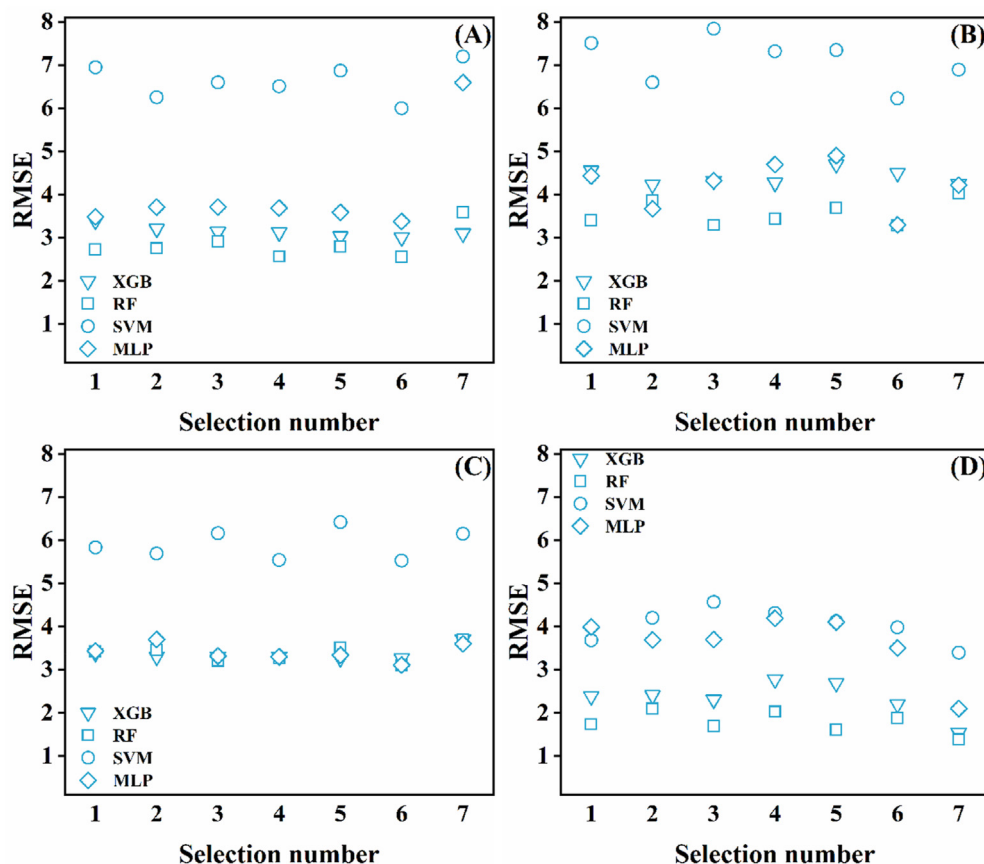


Fig. 4. RMSE of XGB, RF, SVM, and MLP with different input feature selections for the prediction of (A) biochar product, (B) bio-oil product, (C) gas product, and (D) HHV.

**Table 2**  
RF model accuracy information.

| Number      | 1      | 2      | 3      | 4      | 5      | 6      | 7      |
|-------------|--------|--------|--------|--------|--------|--------|--------|
| char_ $R^2$ | 0.9409 | 0.9396 | 0.9326 | 0.9376 | 0.9378 | 0.9479 | 0.8973 |
| oil_ $R^2$  | 0.9412 | 0.9240 | 0.9449 | 0.9401 | 0.9449 | 0.9545 | 0.9175 |
| gas_ $R^2$  | 0.9134 | 0.9115 | 0.9239 | 0.9208 | 0.9085 | 0.9284 | 0.8986 |
| HHV_ $R^2$  | 0.8884 | 0.8380 | 0.8945 | 0.8481 | 0.9048 | 0.8690 | 0.9292 |
| char_RMSE   | 2.7251 | 2.7543 | 2.9158 | 2.5661 | 2.7956 | 2.5572 | 3.5922 |
| oil_RMSE    | 3.4018 | 3.8867 | 3.2923 | 3.4325 | 3.6920 | 3.2823 | 4.0290 |
| gas_RMSE    | 3.4200 | 3.4580 | 3.2064 | 3.2709 | 3.5150 | 3.1002 | 3.7015 |
| HHV_RMSE    | 1.7380 | 2.0939 | 1.6894 | 2.0277 | 1.6049 | 1.8827 | 1.3840 |

with the yields of three-phase products compared with the component classification according to proximate analysis and ultimate analysis. Gani et al. [10] compared the pyrolysis behavior of actual biomass and simulated biomass through thermogravimetric analysis and found that cellulose and lignin content in the feedstock is one of the most important parameters affecting pyrolysis characteristics. Zhu et al. [24] found that the structural components of biomass are more significant than the elementary composition in the prediction of biochar. As for the prediction for HHV of bio-oil, the No.7 selection has apparent advantages above other selections. This indicates that element content is a key feedstock characteristic influencing bio-oil HHV. According to Tang et al. [64], the elementary composition of biomass is more significant than the proximate components in the prediction of bio-oil HHV. Choosing the appropriate input feature selections can reduce the model's complexity and improve the accuracy of the model. Different from the kinetic models, which bear certain input features, models based on machine learning can adopt different combinations of input

features according to their performance. Compared with No.1, selection No.6 and No.7 have fewer input features and were used for further discussion about the three-phase product yields and bio-oil HHV prediction, respectively. Some parameters of the model are shown in the Supplementary materials Table S1.

Fig. 5 shows the model performance on the test set to predict the pyrolysis three-phase product yield and the bio-oil HHV. The figure's point indicates the test sample, and the blue line indicates that the expected value is equal to the experimental value ( $y = x$ ). Some test sample points in Fig. 5 fall on the blue line, and the remaining sample points are close to the blue line and evenly distributed on both sides of the line. This indicates that the model has an excellent predictive effect on three-phase product yield and bio-oil HHV. In addition, the running time of the model is ranged from 15 to 17 s. Compared with the numerical simulation of biomass pyrolysis occupying 50–200 s [14–17], the RF model runs much faster.

### 3.3. RF model feature contribution

Fig. 6 shows the relative importance of feedstock characteristics and pyrolysis conditions in the prediction process. The following internal information can be obtained from No. 6 and No. 7, which can benefit the understanding of the fast pyrolysis conversion process. It is obvious and expected that HT is the key factor affecting the yield of all three-phase products. The change of pyrolysis temperature strongly varies the pyrolysis reaction rate and leads to the variation of product distribution [65]. Besides the primary factor HT, different secondary factors are observed for different output variables. For the prediction of the yield of biochar, lignin

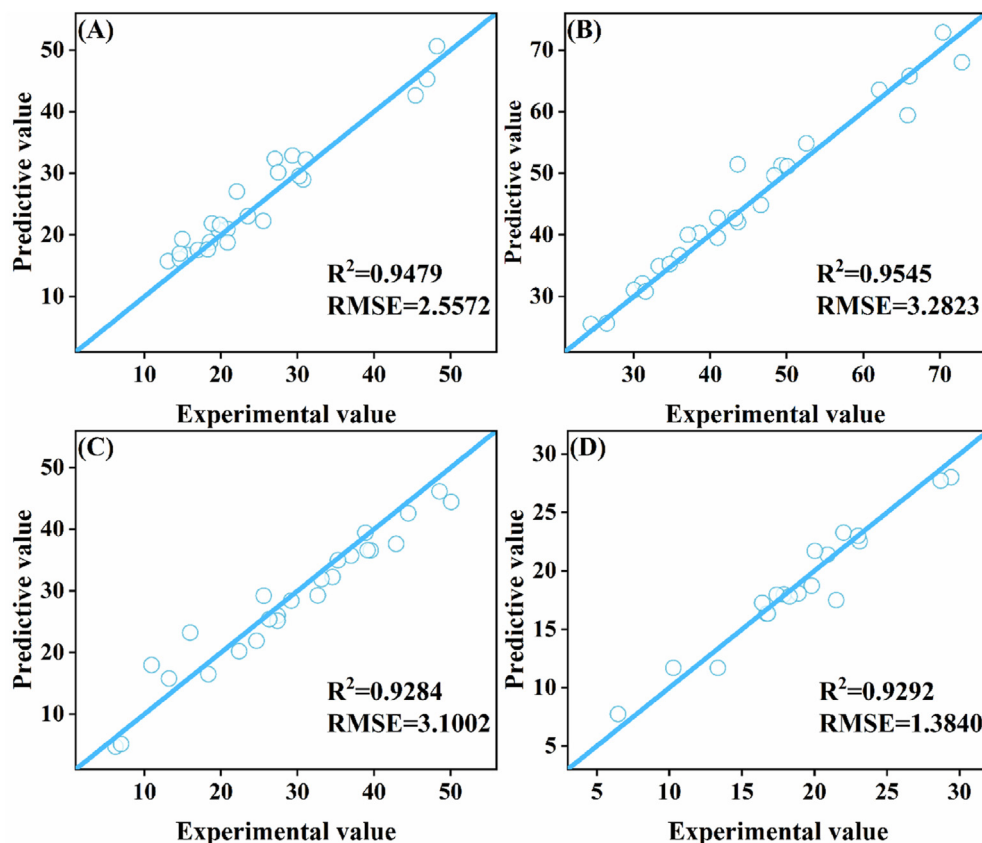


Fig. 5. Comparison of the predicted value of the RF model with experimental values: (A) biochar product, (B) bio-oil product, (C) gas product, (D) HHV.

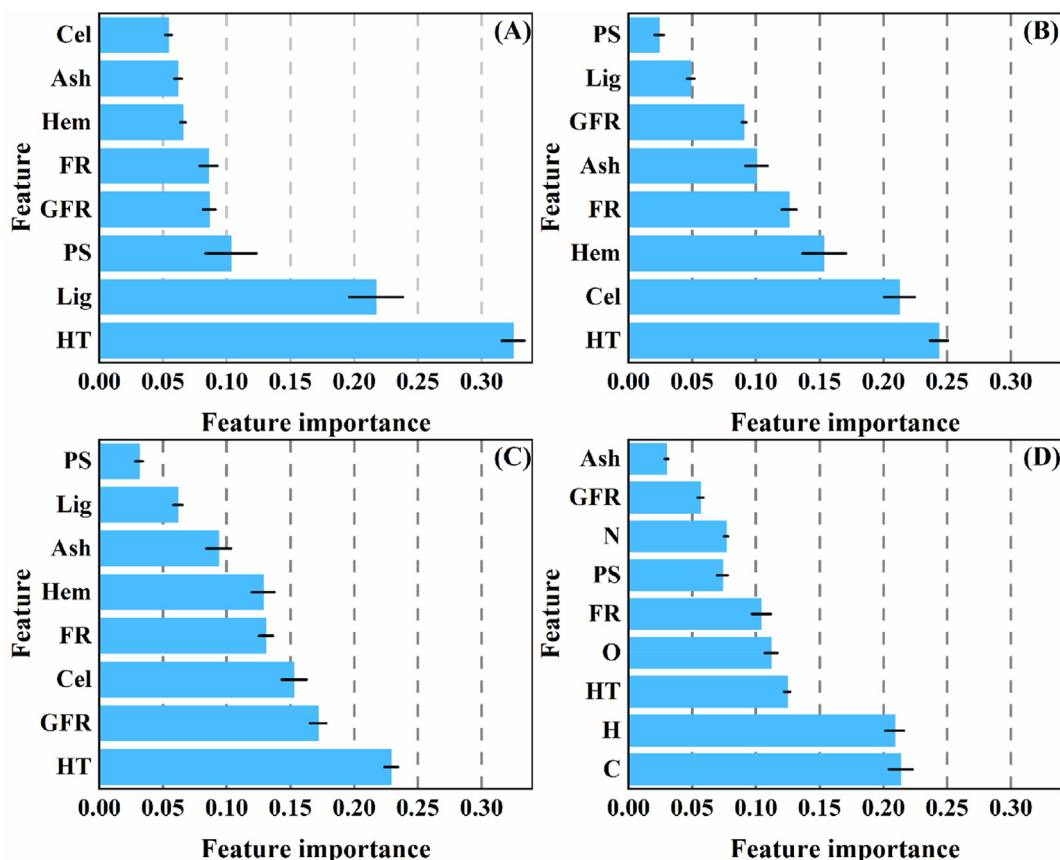
content accounts for the second contribution with obvious advantages above other factors. It was proposed by D. Lathouwers et al. [66] that lignin strongly benefits the formation of char. As for bio-oil yield, the contributions follow the order of Cel > Hem > FR > Ash. Then it indicates that the content of cellulose and hemicellulose can strongly affect the yield of bio-oil, which is attributed to their high yields of bio-oil obtained by pyrolysis [67]. The trace inorganic compound metals K, Na, Ca, Mg, etc., in the ash, are important parameters for the secondary reaction of pyrolysis. Generally, the inorganic compound metal inhibits the formation of bio-oil and promotes the formation of biochar and gas [68]. The limited contribution of ash content to the prediction is perhaps due to the weakened influence of ash content on pyrolysis when the ash content is high. With the increase of ash content, the catalytic effect of ash on biomass pyrolysis becomes saturated. Through pyrolysis experiments, Abdullah et al. [69] found that as the ash content of the pyrolysis raw material increases, the biochar and gas products increase, and the bio-oil yield decreases. However, the slope of the yield change decreases. When the ash content is higher than 5 wt%, the product yield no longer changes with the ash content. In addition, each pyrolysis condition to the bio-oil yield is ranked in descending order of contribution are as follows: pyrolysis temperature, feed rate, GFR, which is consistent with the experimental results of previous literature [42]. When it comes to the gas yield, GFR contributes the most, followed by Cel and FR. The change of GFR can strongly influence the volatile retention time and further alter the degree of secondary reactions in the gas phase, which probably leads to the drastic change of gas yield. Different from the prediction of the three-phase product yields, that of the bio-oil HHV is less affected by the pyrolysis conditions. It is interesting that among the several features of pyrolysis conditions, the

pyrolysis temperature accounts for the most considerable contribution of only 0.1246. Instead, the feedstock characteristics are the key factor to determining the HHV, with the assistance of 0.7164 (Fig. 6D), which is in line with previous work [23]. Through pyrolysis experiments of different crops, Phan et al. [70] found that for rice straw and bagasse with different feedstock characteristics, the H<sub>2</sub>O content in bio-oil strongly varied from 27.94 wt% to 16.73 wt%, and the bio-oil HHV strongly varied and from 17.367 MJ/kg to 19.360 MJ/kg. Instead, when the pyrolysis temperature rose from 470 °C to 515 °C, the water content and HHV of the bio-oil produced by the pyrolysis of rice straw only changed 0.62 wt% and 0.318 MJ/kg, respectively. This indicates that the characteristics of feedstock have a greater influence on the characteristics of bio-oil than pyrolysis conditions. Furthermore, the high contributions of the content of hydrogen and carbon to the bio-oil HHV indicate the crucial effects of the two elements in feedstock on bio-oil quality, which confirms the importance of pretreatment such as torrefaction before pyrolysis. In addition, although a strong effect of ash has been observed on biomass pyrolysis [71], its contribution to the production of three-phase products and the bio-oil HHV is relatively low.

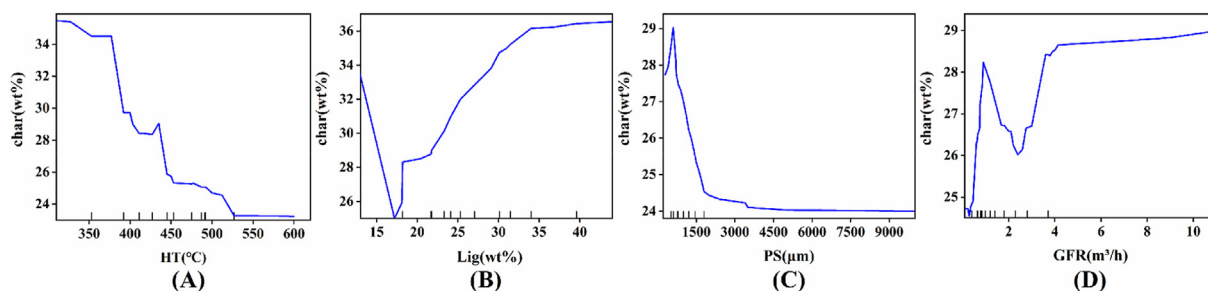
### 3.4. RF model partial dependency analysis

#### 3.4.1. Partial dependence plot (PDP) analysis for biochar yield

The PDP analysis only considers the dependency relationship between one or two input features and the target output feature, while other features are input with average values, thus marginalizing other features. Through the analysis, the specific effects of feedstock characteristics and pyrolysis conditions on pyrolysis products can be revealed. Fig. 7 shows the partial dependence of



**Fig. 6.** The contribution of different input features to the prediction of three-phase product yield and bio-oil HHV: (A) biochar product, (B) bio-oil product, (C) gas product, (D) bio-oil HHV.



**Fig. 7.** PDP of biochar yield on input feature (A) HT, (B) Lig, (C) PS and (D) GFR.

biochar yield on a single variable. The scale line above the x-axis indicates the density of data, and the trend line is not accurate when the scale line is too sparse. So it is necessary to combine the distribution box plot of each feature shown in Fig. S1 and Fig. S2 to analyze the results. As shown in Fig. 7A, the influence of temperatures on the biochar yield showed a nearly linear decrease, but the slopes decreased gradually with the increase of pyrolysis temperature. Through the statistical analysis results of the biochar literature data set, it is found that the pyrolysis temperature has a significant negative correlation with the yield ( $p < 0.05$ ) [72]. When the temperature is higher than 450 °C, the slope becomes so small. It can be seen that the pyrolysis of the three components of biomass tends to be completed at 450 °C [73]. In the concentrated distribution range of lignin content of 21–34 wt %, the char yield increases with lignin content. Yang et al. [74] conducted the pyrolysis experiments of biomass feedstocks with different lignin contents.

The results show that the yield of pyrolysis char increased with higher lignin contents. As for the particle size of concentrated 600–2000 μm, with increasing particle size, the char yield increases for PS region of 600–800 μm but decreases for that of 800–2000 μm. The increase of char yield at elevated particle size of 600–800 μm is consistent with previous work suggesting a promotion effect of large particle size on char formation [65,75–79]. These are often attributed to the promoted interactions between char and volatiles and decreased actual heating rates by larger particle size. Instead, when the particle size is above 800 μm, the change trend is reversed. Xiong et al. [77] conducted the modeling of biomass fast pyrolysis in bubbling fluidized bed reactors. It was suggested that provided the effects of intraparticle temperature gradient on the reaction kinetics are ignored, the increase in biomass particle diameter increases the particle Reynolds number, makes it more difficult for biomass particles to be blown out, and



provides a longer residence time for the conversion of biomass. As a result, the yield of the so-called char, which also includes unreacted biomass, is decreased. Considering the complex effects of particle size on biomass pyrolysis, further work is still needed. As for the GFR shown in Fig. 7D, when it is in the concentrated region (0.6–2.6 m<sup>3</sup>/h), the biochar yield first increases and then decreases with the increase of the GFR. The non-monotonic variation of char yield with increasing GFR is due to the competition between the decreased decomposition of unreacted biomass and the promoted removal of tar instead of condensation reaction forming char and gas.

Fig. 8 shows the partial dependence of char yield on two input features in the concentrated regions and the interactions between them. Lignin and pyrolysis temperature have no interactive effect on the yield of biochar (Fig. 8A). No obvious synergistic effect of the two features is observed, which is expected considering the even decomposition of lignin at elevated temperature [67]. As shown in Fig. 8B, small raw material particle size and low pyrolysis temperature synergistically promote the production of biochar. Bearing the conclusion by Xiong et al. [77] mentioned above, it is plausible that small particle size and low heating temperature synergistically suppress the conversion of biomass and lead to the highest char yield. Furthermore, as shown in Fig. 8E, a synergistic promotion effect is observed of high lignin content and low particle size on char formation, which indicates that lignin accounts for most of the unreacted biomass due to its higher stability compared with cellulose and hemicellulose [67]. As for GFR in Fig. 8C, D, and F, the range of 1–1.2 m<sup>3</sup>/h is beneficial to increase the yield of biochar.

### 3.4.2. PDP analysis for bio-oil yield

Bearing in mind that HT, the content of cellulose and hemicellulose and FR are the four dominant factors for the prediction of bio-oil, Fig. 9 shows the partial dependence of bio-oil yield on these input features. At temperatures less than 450 °C, the yield of oil

increases roughly linearly as the temperature rise, while at temperatures above 480 °C, it decreases linearly with the pyrolysis temperature. The highest yield of oil is obtained at 450–480 °C (Fig. 9A), which is consistent with previous pyrolysis experiments [42]. As the temperature increases, the breaking of chemical bonds in the biomass is accelerated, which is beneficial to increase the yield of bio-oil. But at very high temperatures, the volatile matter is secondarily cracked into biochar and gas, thereby reducing the bio-oil [80]. The content distribution ranges of cellulose and hemicellulose are concentrated in 32–46 wt % and 15–30 wt %, respectively, as shown in Fig. S1B. Generally, as the cellulose and hemicellulose contents increase, the yield of oil increases linearly with a large slope, as shown in Fig. 9B and C. This is in line with the experimental results that the pyrolysis of cellulose and hemicellulose yielded more liquid products than lignin [67]. Different from lignin with relatively stable benzene rings preferring to char formation, cellulose, and hemicellulose with unstable sugar rings and reactive hydroxyls are easy to break into liquid products [73]. According to Fig. S1E, the FR is generally less than 1 kg/h, and the bio-oil yield basically decreases linearly with the increase in feed rate and has a large slope (Fig. 9D), which is attributed to two mechanisms. First is the suppressed removal of primary tar leading to the decrease of final tar yield through secondary reactions. Second is the insufficient heat provided for biomass conversion in a limited time at high feed rate [77].

Since the primary decomposition temperature of cellulose and hemicellulose are about 315–400 °C and 220–315 °C, respectively, it can be understood that no obvious synergy effect is observed of the contents of the two components and heating temperature above 430 °C as shown in Fig. 10A and B. In Fig. 10C, it is obvious that low flow rate with low heat demand reinforces the effect of heating temperature on bio-oil yield, resulting in a significant high yield of bio-oil at about 450 °C and the lowest feed rate. Instead, as shown in Fig. 10D and F, limited synergy effect is presented of the

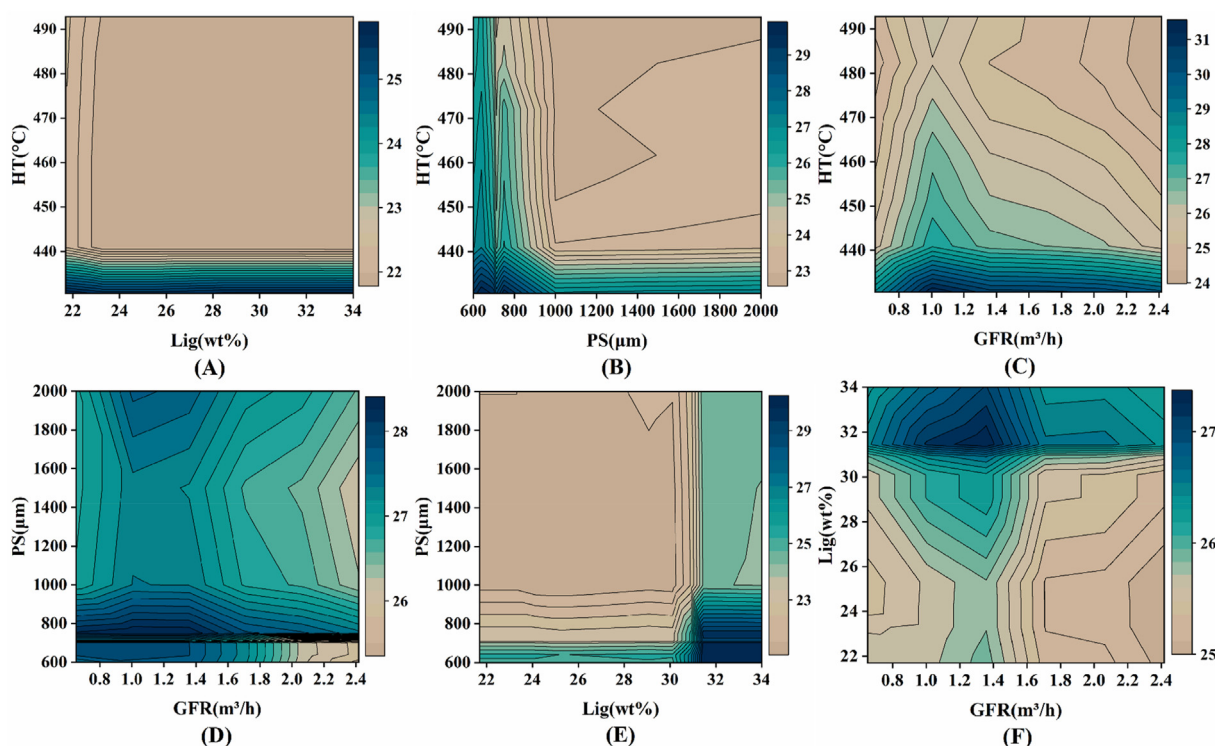


Fig. 8. The PDP of biochar yields on two input variables.

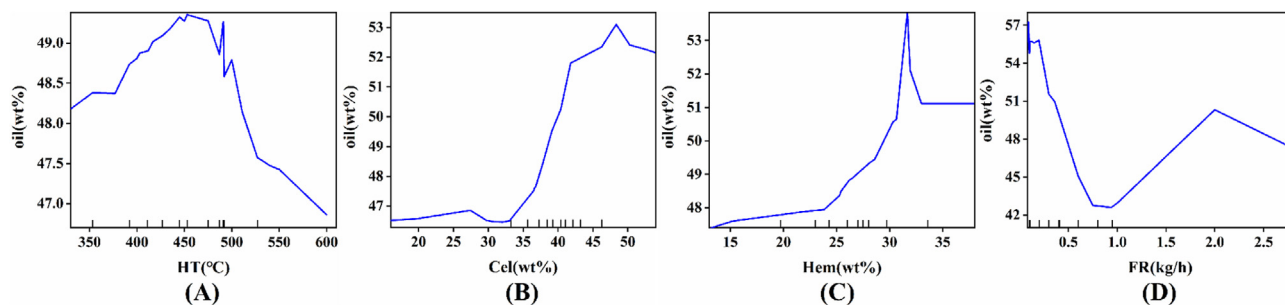


Fig. 9. PDP of bio-oil yield on input feature (A) HT, (B) Cel, (C) Hem and (D) FR.

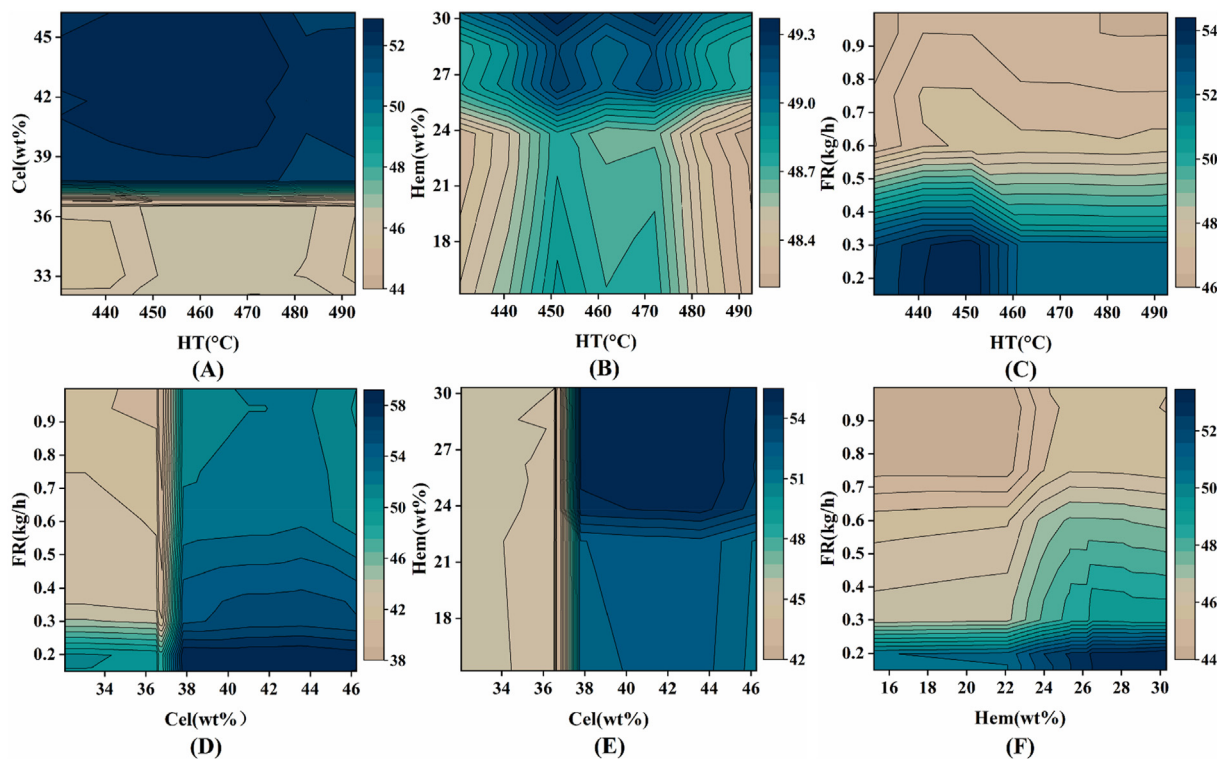


Fig. 10. The PDP of bio-oil yield on two input variables.

contents of the two components and FR, which indicates that the effect of FR on bio-oil yield is more probably due to the change of heat transfer conditions [77]. In addition, Fig. 10E presents that higher content of cellulose obviously enhances the promotion effect of hemicellulose on the formation of bio-oil, which is in line with previous work by Wu et al. [81] suggesting that cellulose-hemicellulose interaction would significantly promote the formation of hemicellulose-derived products.

### 3.4.3. PDP analysis for gas yield

As shown in Fig. 11A, as the temperature increases, the gas production increases roughly linearly at the expense of bio-oil and char, as discussed above. This is probably due to the accelerated secondary cracking of primary volatiles into gases at high temperatures [42]. Fig. 11B shows the effect of GFR on gas formation. The same change trend is observed as that of char yield with increasing GFR. As for the cellulose content ranging from 32 wt % to 46 wt %, the formation of gas products is strongly inhibited by high cellulose content, which promotes the formation of bio-oil as mentioned above. When the feed rate is less than 1 kg/h, the gas

production rate generally decreases with the increase of the feed rate (Fig. 11D). The faster feed rate is not conducive to the heating of the feedstock, hinders the precipitation of volatiles, and reduces the gas yield. Fig. 12 shows the partial dependence of gas yield on two input features in the concentrated regions and the interactions between them. For most of the combinations of two input features, no obvious synergy effect is observed except FR and the content of cellulose. Obviously, a lower content of cellulose enhances the promotion effect of high FR on gas formation. This indicates that more attentions are needed to control the FR for the fast pyrolysis of biomass with less cellulose.

### 3.4.4. PDP analysis for HHV

As mentioned above, different from the yields of three-phase products strongly influenced by pyrolysis conditions, the bio-oil HHV is more sensitive to the feedstock characteristics. As shown in Fig. 13, in the concentration range of carbon content of 44–50 wt % and the concentration range of hydrogen content of 5–6 wt %, the HHV increases roughly linearly with the increase of the contents of two elements. The higher the carbon content in the

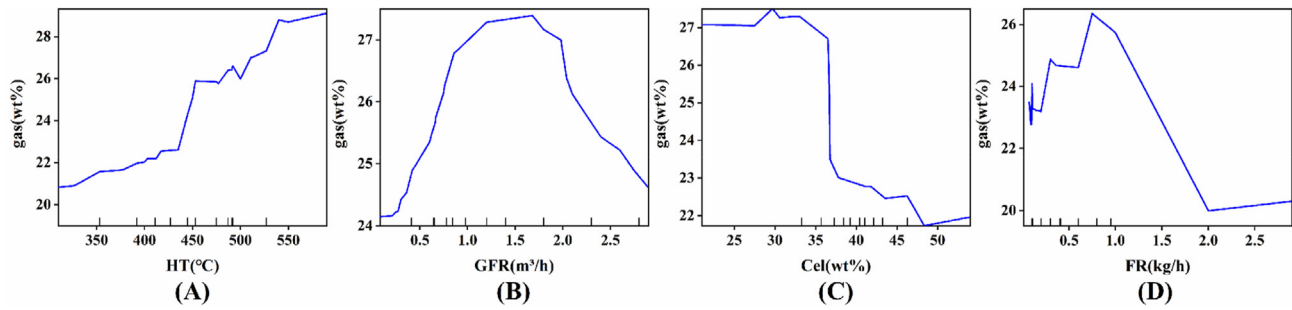


Fig. 11. PDP of gas yield on input feature (A) HT, (B) GFR, (C) Cel and (D) FR.

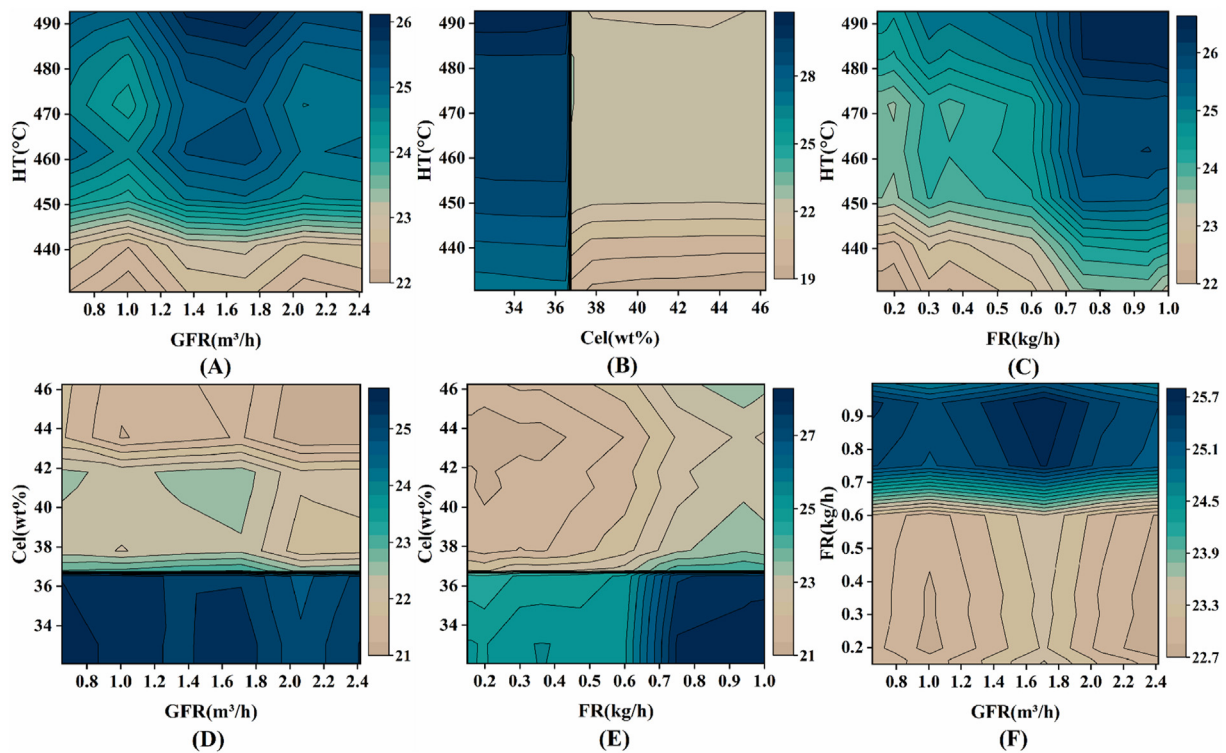


Fig. 12. The PDP of gas yield on two input variables.

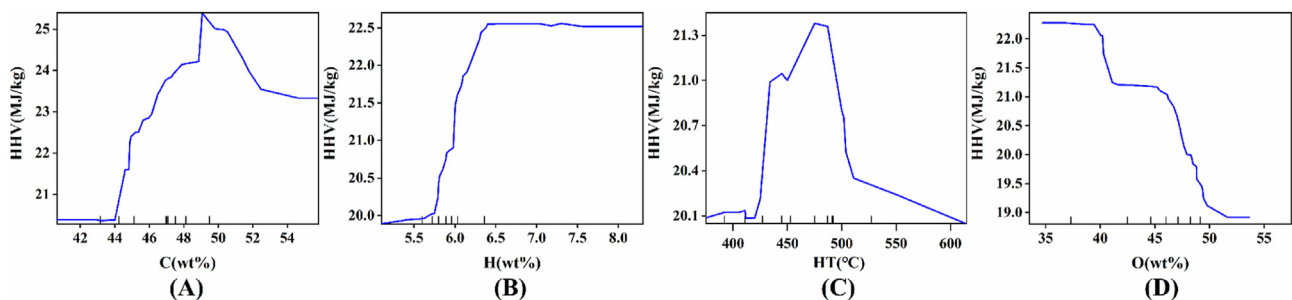


Fig. 13. PDP of HHV on input feature (A) C, (B) H, (C) HT and (D) O.

feedstock, the lower the O/C ratio of the pyrolyzed bio-oil, which is beneficial to increase the bio-oil HHV. Tang et al. [64] established a bio-oil hydrogen content prediction model and found that the hydrogen content in the bio-oil increases as the feedstock hydrogen content increases, and the higher the hydrogen content in the bio-oil, the better the quality of the bio-oil. In contrast, high oxygen

content in the feedstock is unfavorable to the enhancement of bio-oil HHV. Then it can be concluded that pretreatment of feedstock such as torrefaction aiming at deoxygenation is of crucial importance to the quality improvement of bio-oil. Besides the feedstock characteristic, pyrolysis temperature still performs an obvious influence on bio-oil HHV. The highest value is obtained at about

480 °C, which is close to the temperature of the highest yield of bio-oil. Pyrolysis experiments of sawdust by Salehi et al. [39] show that when the pyrolysis temperature increased from 425 °C to 550 °C, the oxygen content of bio-oil first decreased and then increased. The high oxygen content in the bio-oil leads to a high content of acids, ethers, esters, and other high-oxygen substances, which makes the bio-oil acidic and reduces the bio-oil HHV. Then it can be concluded that 480 °C is the optimum pyrolysis temperature of fast pyrolysis of biomass in a bubbling fluidized bed. However, compared with the pyrolysis temperature, the characteristics of feedstock, especially the elementary composition, perform a stronger influence on bio-oil HHV. Besides the simple transfer of differences in elementary composition from feedstock to bio-oil during biomass pyrolysis, the interaction of carbon, hydrogen, and oxygen in the feedstock also makes a certain contribution. According to work by Pütün et al. [82], the higher hydrogen contents of feedstock lead to more opportunities to form aromaticity, aromatic ring-joining methylene, and naphthenic protons, which weakens the influence of oxygen on bio-oil and increases the bio-oil HHV. This can be supported by the results shown in Fig. 6D. Due to the interaction, hydrogen and carbon contents make more contributions to bio-oil HHV than oxygen content. As for PDP on two features shown in Fig. 14, no obvious interactions can be observed between all the combinations.

#### 4. Conclusions

Taking comprehensive feedstock characteristics and pyrolysis conditions as input, three-phase product yield, and bio-oil HHV as output, regression models were established based on XGB, RF, SVM, and MLP algorithms. The models were optimized in terms of model feature input selection and hyperparameter adjustment. Seven different input feature combinations were considered and compared for different algorithms based on the evaluation of prediction accuracy. The results show that the RF model is most

suitable for the prediction of three-phase product yields and bio-oil HHV. Furthermore, the combination of input features including Ash, Cel, Hem, Lig, PS, HT, GFR and FR is suitable for predicting product yields, and that including Ash, C, H, O, N, PS, HT, GFR and FR is suitable for HHV prediction. Under the above input characteristics, the  $R^2$  and RMSE of the random forest model for predicting char, oil, gas, and the HHV are 0.9479, 0.9545, 0.9284, 0.9292, and 2.5572, 3.2823, 3.1002, 1.3840, respectively.

The sub-models internal information by visual analysis presents that HT is the most critical factor affecting three-phase product distribution. Instead, bio-oil HHV is more affected by the feedstock characteristics with the contents of C and H as the critical factors. Through PDP analysis, the impact way of the influential factors and the interactions between them were obtained. The highest yield and HHV of bio-oil is obtained at HT of about 480 °C, suggesting 480 °C as the optimum pyrolysis temperature of fast pyrolysis of biomass in a bubbling fluidized bed. As for the feedstock characteristics, compared with cellulose and hemicellulose mostly benefiting tar formation, lignin is mainly responsible for char formation. Furthermore, the interaction between cellulose and hemicellulose is observed, which promotes tar formation. High contents of C and H and low content of O is favorable to the enhancement of bio-oil HHV, indicating that pretreatment of feedstock such as torrefaction aiming at deoxygenation is of crucial importance to the quality improvement of bio-oil.

#### Credit author statement

Erwei Leng: Writing - original draft, Writing - review & editing, Funding acquisition. Ben He: Software, Data curation. Jingwei Chen: Conceptualization. Gaoliang Liao: Visualization. Yinjie Ma: Visualization. Feng Zhang: Visualization. Shuai Liu: Resources. Jiaqiang E: Project administration.

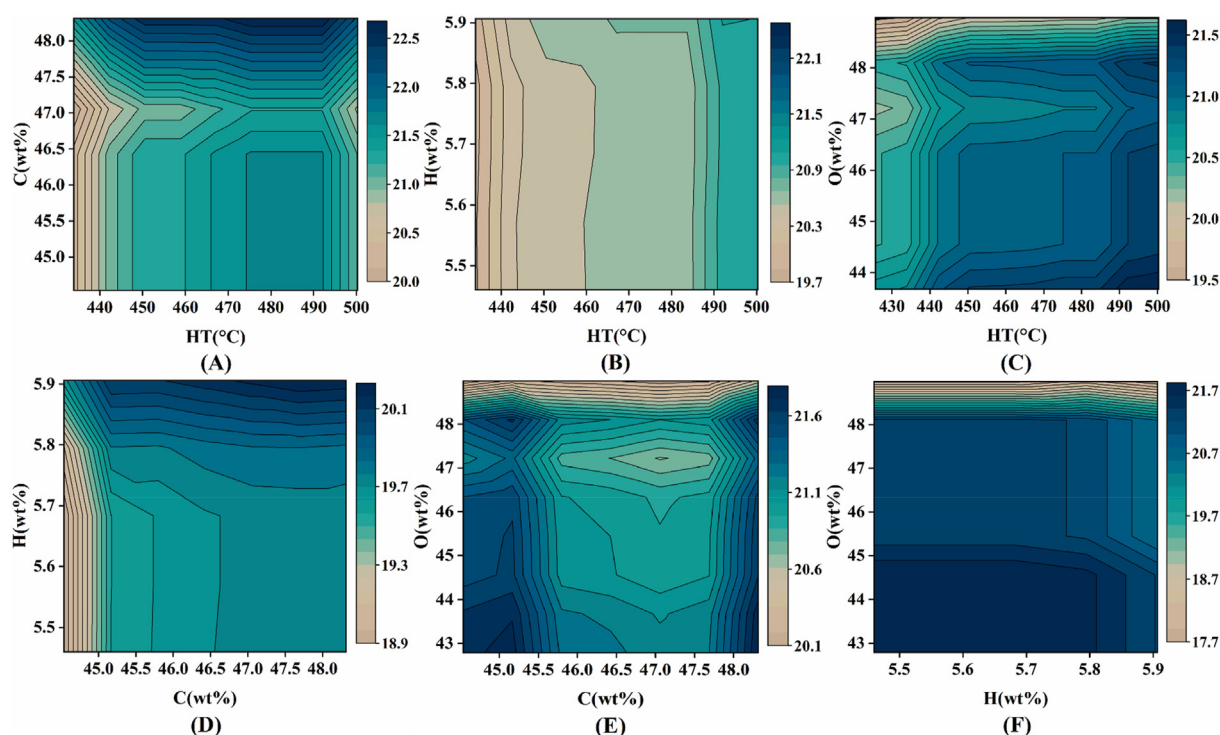


Fig. 14. The PDP of HHV on two input variables.



## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

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